Model Performance Review

Coverage

- **Model outputs reviewed:** `OLS`, `OLS3`, `ENET`, `PLS`, `PCR`
- **Prediction horizon reviewed:** 180 monthly out-of-sample periods from **2010-08-31** to **2025-07-31**
- **Full-sample prediction rows per model:** 34,515
- **Common artifact families used:** `r2`, `portfolio`, `benchmark`, `predictions`, `windows`, `importance`, `complexity`

Key Takeaways

- **Best predictor on the headline squared-error metric:** `ENET`
- **Best investable long-short model:** `ENET`
- **Best long-only model:** `OLS`
- **Most stable model across rolling windows:** `ENET`
- **Weakest completed model in this batch:** `OLS3`

My takes

1. `ENET` is the strongest completed model overall. It has the best full-sample out-of-sample `R²`, the strongest long-short net Sharpe after 30 bps cost, the highest benchmark information ratio, and the best cumulative long-short path.
2. `OLS` is not the best predictor on `R²`, but it remains a strong long-only model. Its long-only net Sharpe is the highest in this completed batch even though its `R²` is negative. That means portfolio payoff and point-forecast fit are not the same thing in these runs.
3. `PLS` and `PCR` are credible middle-ground models. They do not beat `ENET`, but both deliver usable long-short Sharpe above 1 and materially better results than `OLS3`.
4. `OLS3` is materially weaker than the richer linear alternatives. In this dataset, the 3-factor restricted specification is not capturing enough cross-sectional variation to compete.
5. The model ranking changes depending on the objective. If the goal is **prediction fit**, `ENET` is best. If the goal is **long-only portfolio quality**, `OLS` is strongest among completed runs. If the goal is **best overall tradeable signal**, `ENET` is the best current choice.

Quick Comparison Table

Model	R2 OOS Full	R2 OOS Large	R2 OOS Small	Mean Rank IC	Long-only Sharpe	Long-short Sharpe	Benchmark IR
ENET	0.0044	0.0039	-0.0003	0.0785	0.9705	1.6960	0.2467
OLS	-0.0143	-0.0588	-0.0326	0.1044	1.2651	1.3013	0.0341
PCR	0.0025	0.0004	-0.0014	0.0545	0.8250	1.1464	0.0976
PLS	0.0008	-0.0019	-0.0034	0.0724	0.8423	1.1394	0.1408
OLS3	-0.0113	-0.0116	-0.0124	0.0306	0.3251	-0.0542	-0.2443

Quick Notations and Formulas:

1. Information coeficients (IC)

$$IC_t = \mathrm{corr}(\hat{y}_t,\ y_t)$$

where for each date t:

- $\hat{y}_t$ = model scores or predicted returns across stocks,
- y_t = realized returns across the same stocks.
  Then:

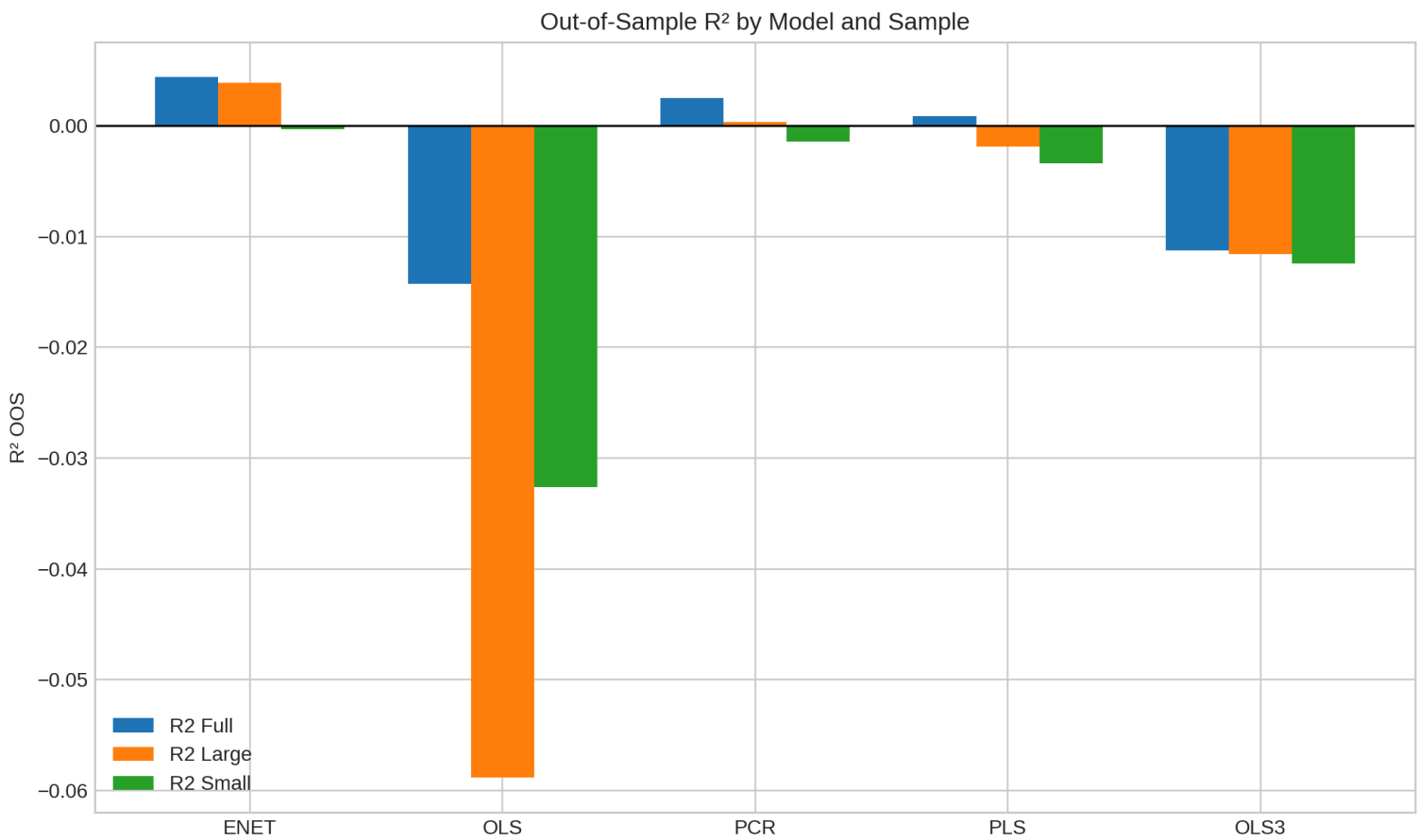
$$\text{Mean IC} = \frac{1}{T}\sum_{t=1}^{T} IC_t$$

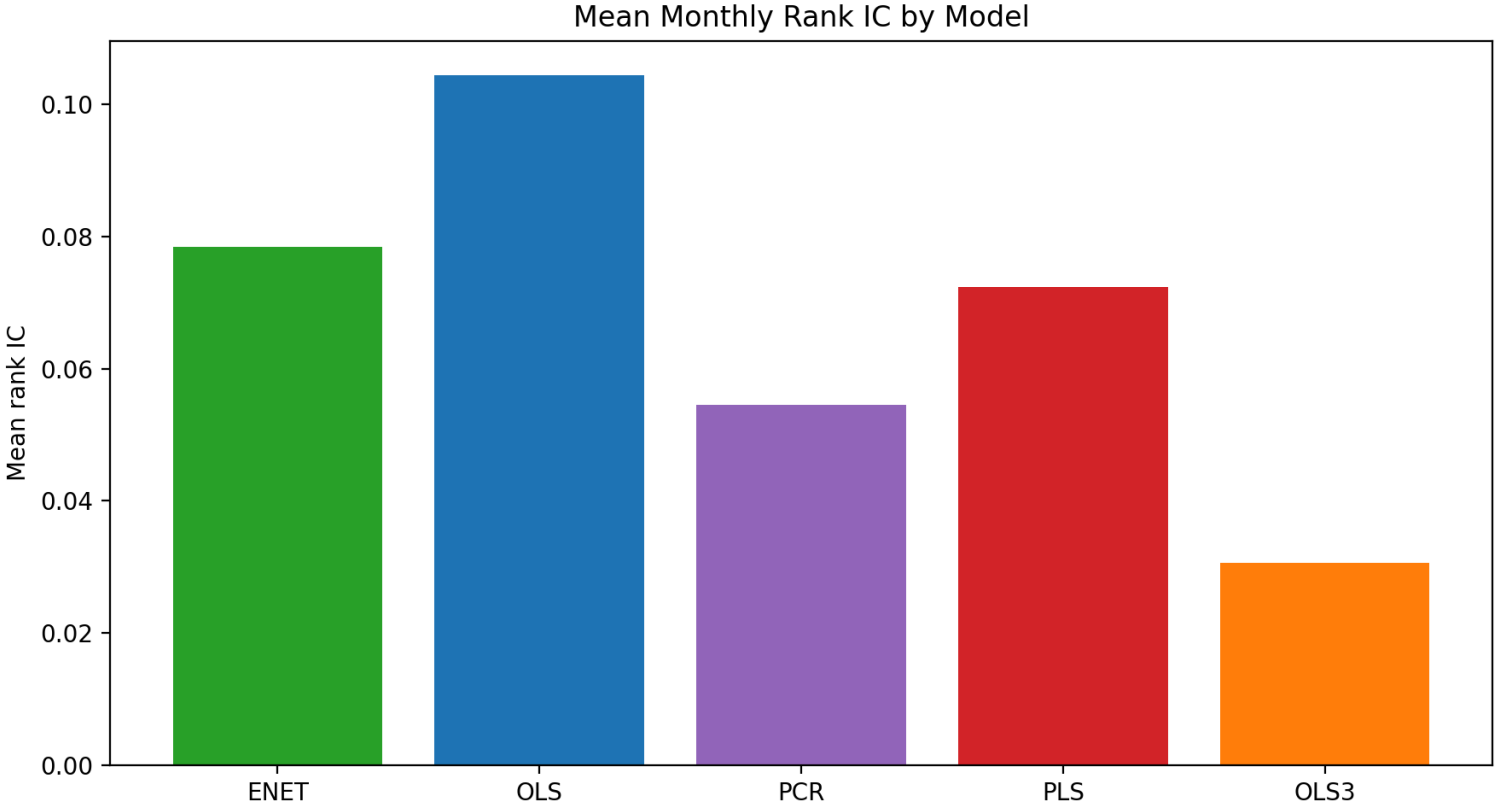
1. Information Ratio (IR)
   Benchmark IR (information ratio) is calculated as:

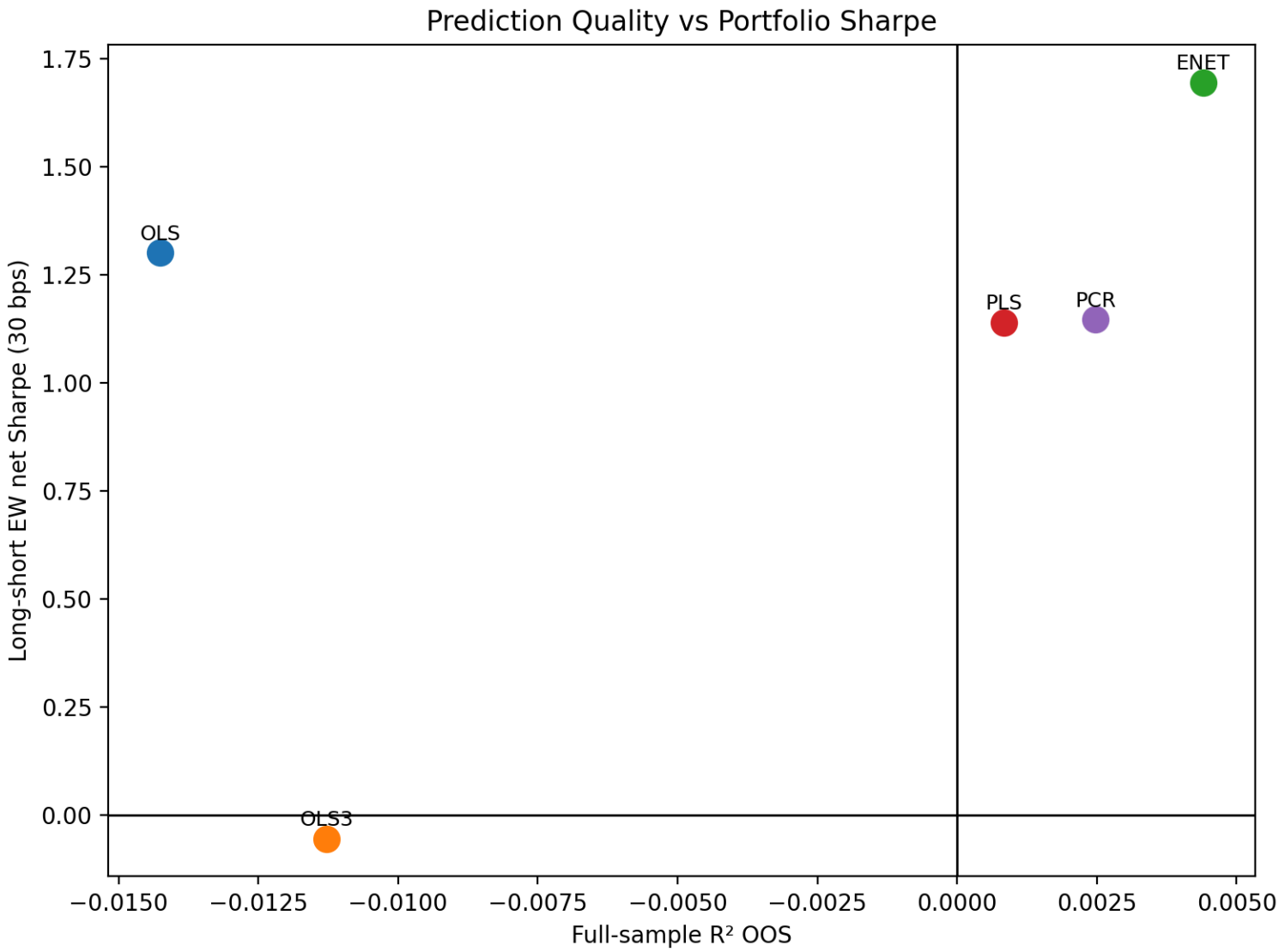
$$AR_t = R_t^{strat} - R_t^{bench}, \quad IR = \frac{\overline{AR}}{\sigma(AR)}.$$

R_t^{bench} here I take the VN30 index (a portfolio with 30 biggest market capitalization at at the time of the test period, and re-balance every 6 month, equally weighted)

Prediction Quality




Mean Monthly Rank IC by Model
Mean rank IC
0.00
0.02
0.04
0.06
0.08
0.10
ENET
OLS
PCR
PLS
OLS3


Prediction Quality vs Portfolio Sharpe
Long-short EW net Sharpe (30 bps)
Full-sample R² OOS
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
−0.0150
−0.0125
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
ENET
OLS
PLS
PCR
OLS3

Prediction metrics table

Model	R2 OOS Full	Pearson(y,yhat)	Spearman(y,yhat)	Mean Rank IC	Share Positive IC	Sign Accuracy
ENET	0.0044	0.0585	0.0535	0.0785	71.7%	51.5%
OLS	-0.0143	0.0629	0.0850	0.1044	76.7%	53.6%
PCR	0.0025	0.0529	0.0515	0.0545	68.9%	51.1%
PLS	0.0008	0.0551	0.0549	0.0724	70.0%	51.4%
OLS3	-0.0113	0.0182	0.0350	0.0306	56.7%	52.7%

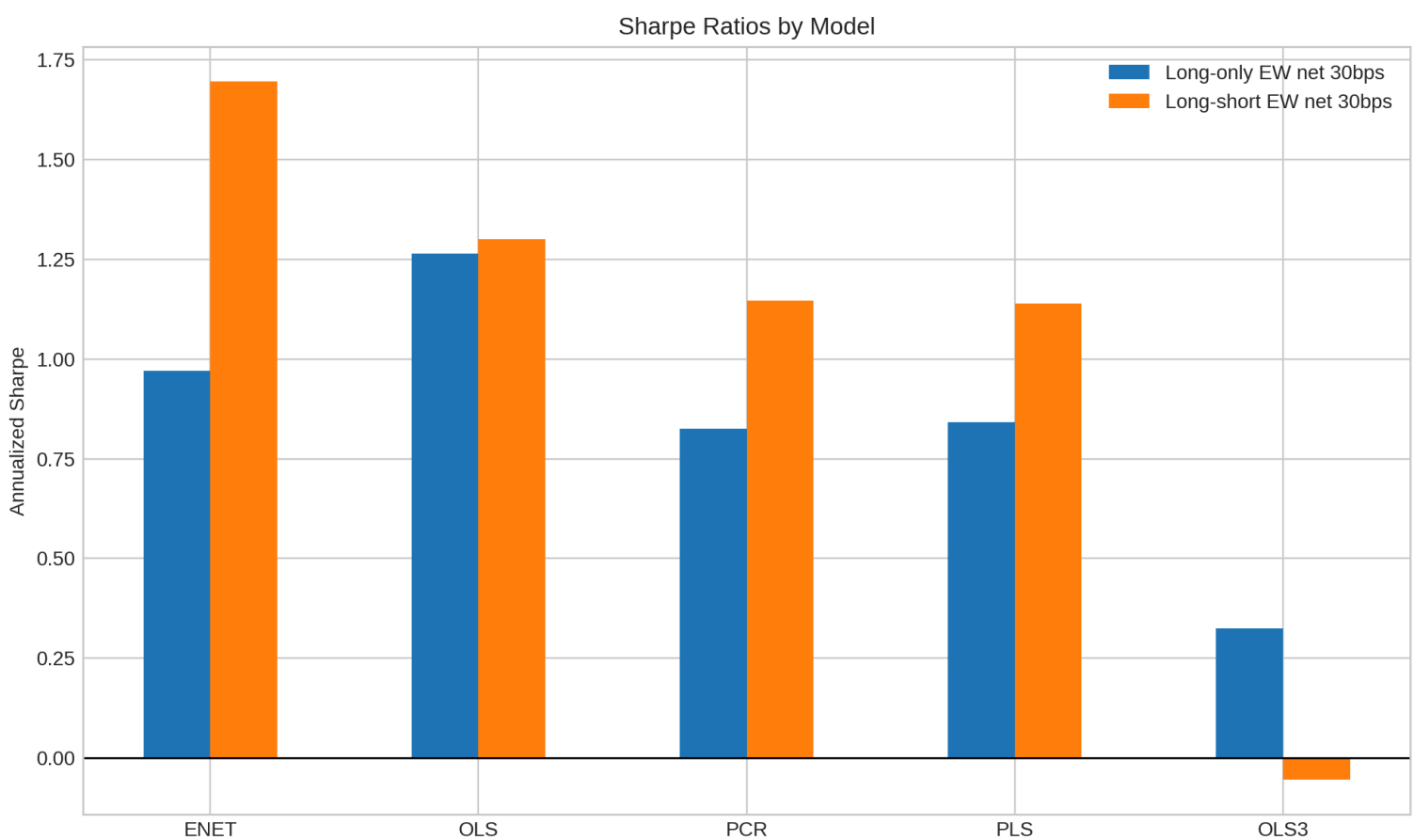
Interpretation

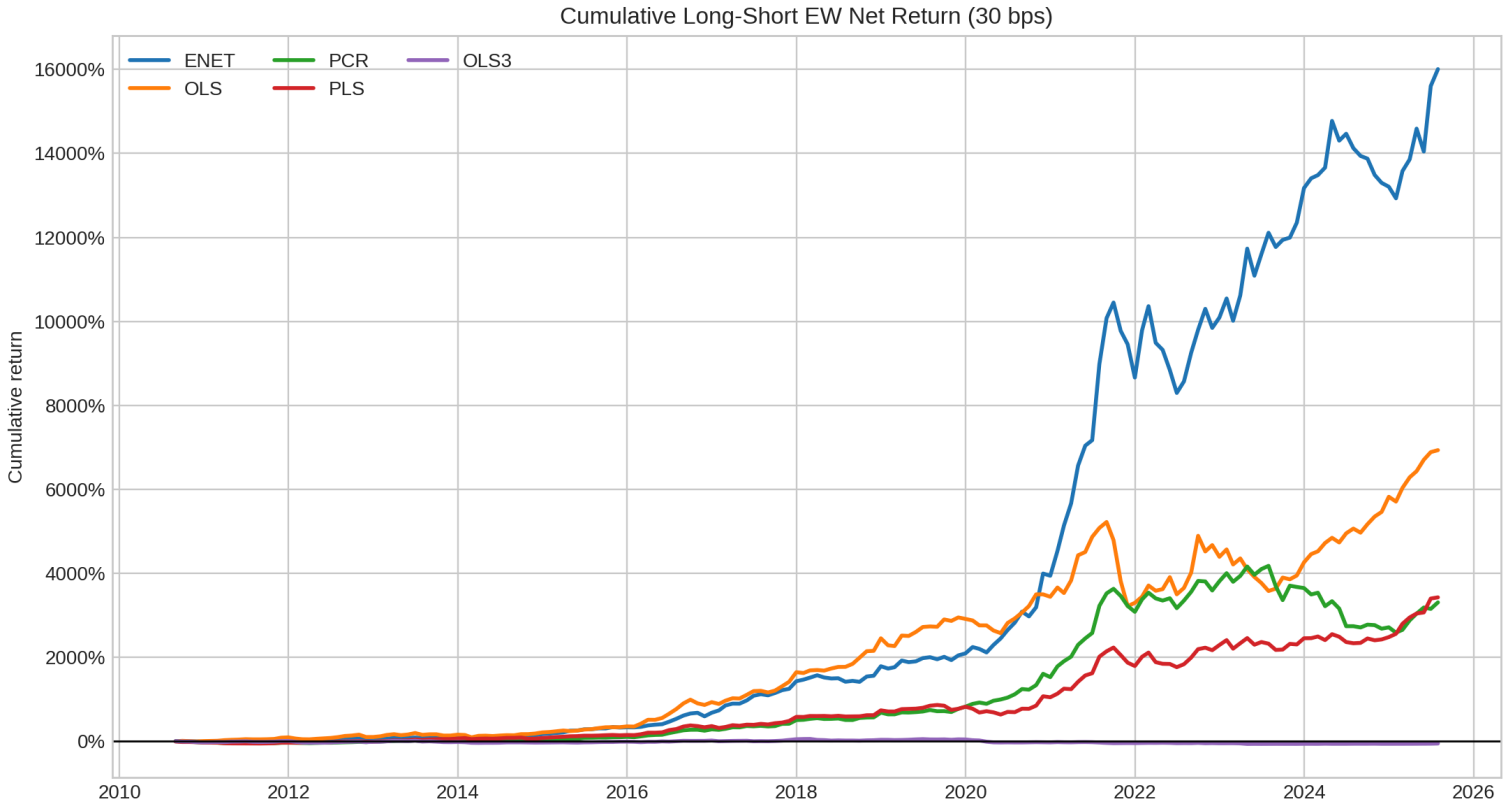
- `ENET` is the only completed model with clearly positive `R²` in both the **full** and **large-firm** samples.
- `PCR` is also positive on full and large samples, but its `R²` is weaker than `ENET`.
- `PLS` is barely positive on full sample and negative on large/small subsamples.
- `OLS` has the **highest mean rank IC** and **highest sign accuracy**, but still negative `R²`. That is a concrete example of why portfolio sorting metrics and squared-error fit can disagree.
- `OLS3` is weak on every predictive dimension reviewed here.

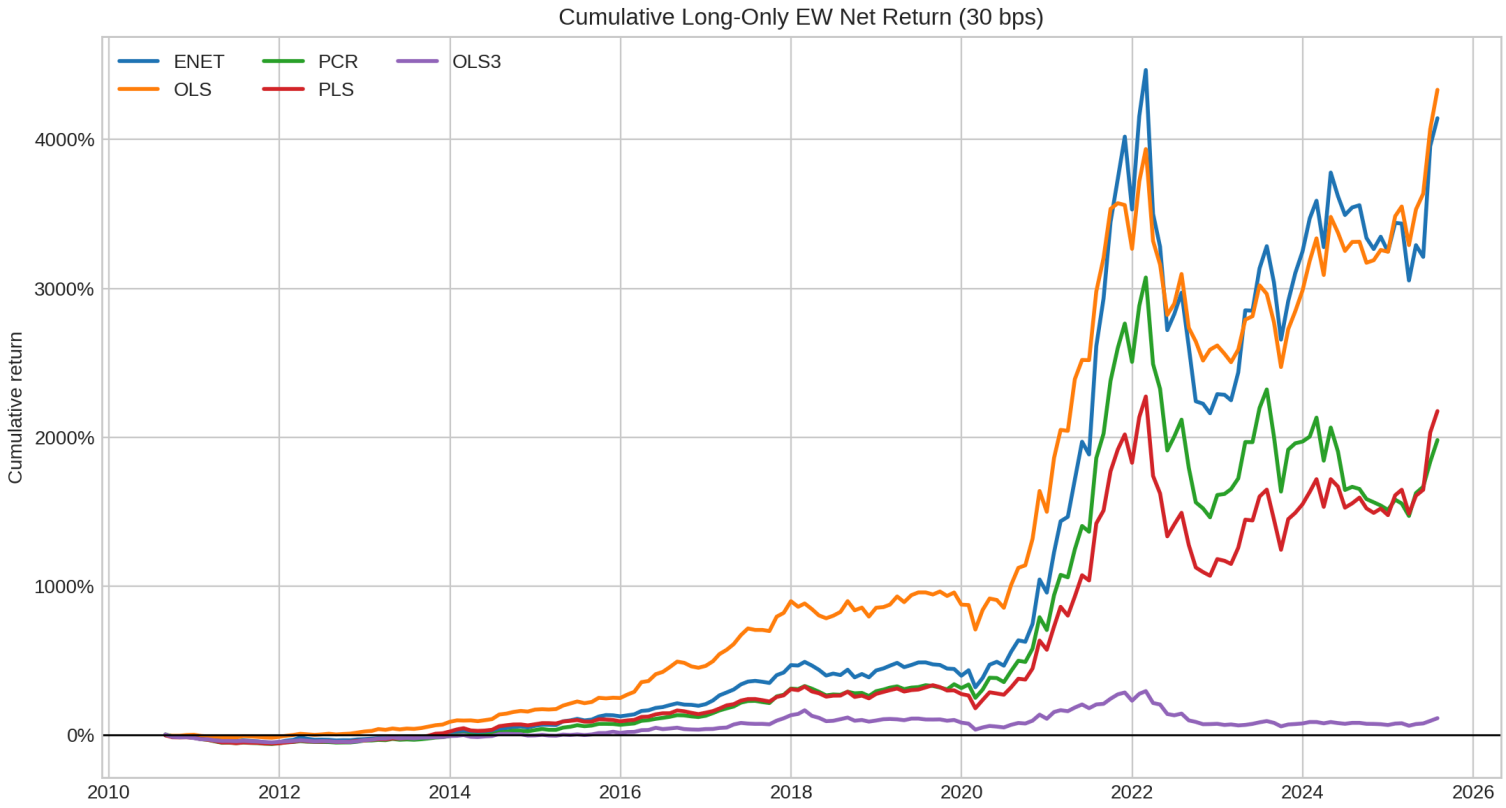
These results so far tell me

- "Predict the best" as **highest out-of-sample** **`R²`**, the answer is `ENET`.
- **Best monthly ranking ability**, `OLS` looks better than its `R²` suggests.
- The safest interpretation is that `ENET` is the best balanced predictor, while `OLS` is a noisier but sometimes economically stronger ranking signal.

Portfolio Performance







Net 30 bps portfolio summary

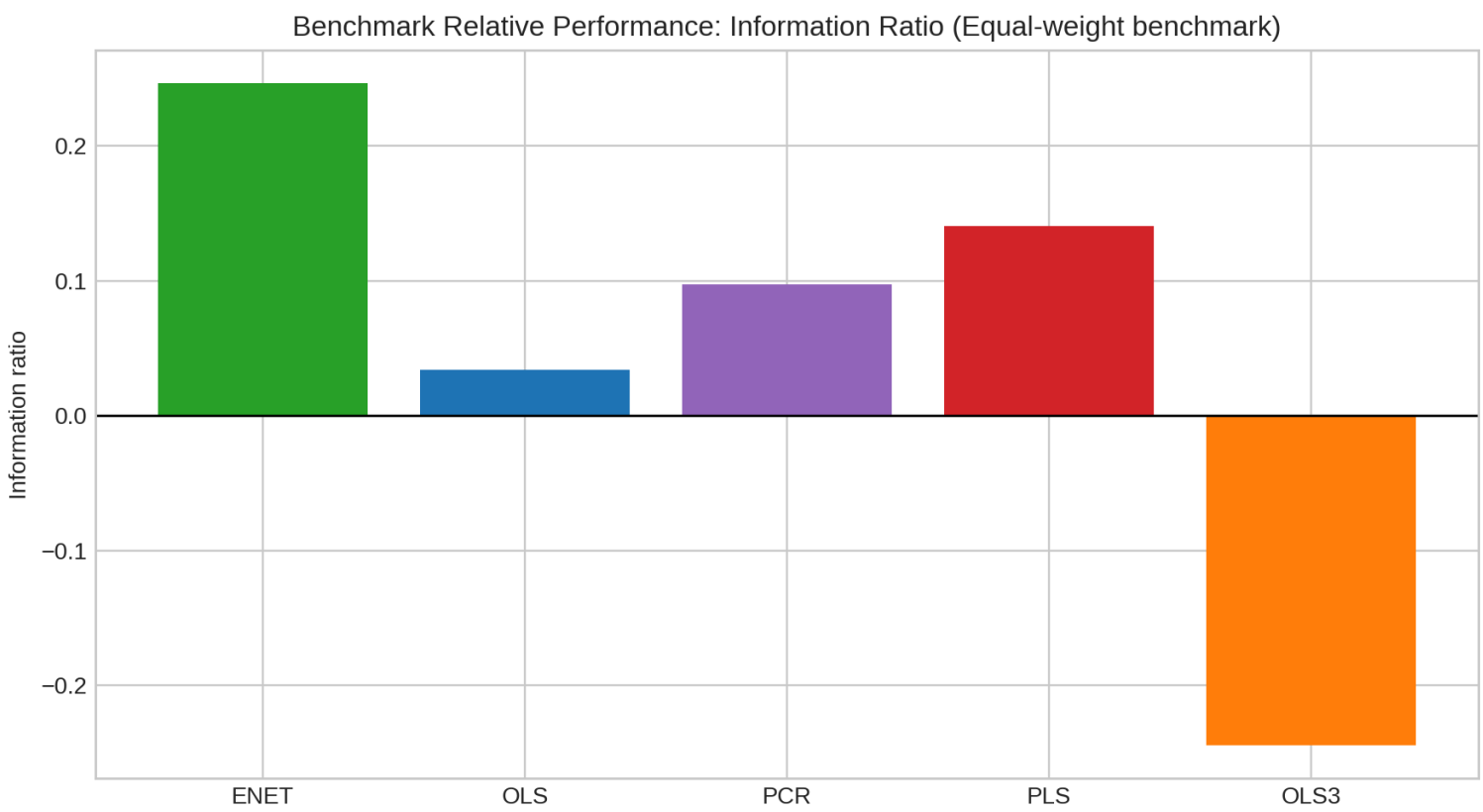
Model	Long Ann Ret	Long Sharpe	Long Cum Ret (x)	Long Max DD	LS Ann Ret	LS Sharpe	LS Cum Ret (x)	LS Max DD	LS Mean Turnover
ENET	28.4%	0.970	41.434	-51.0%	40.3%	1.696	160.073	-26.2%	1.330
OLS	28.8%	1.265	43.343	-36.3%	32.8%	1.301	69.351	-37.6%	1.639
PCR	22.4%	0.825	19.805	-63.1%	26.5%	1.146	33.100	-50.1%	1.219
PLS	23.2%	0.842	21.752	-57.5%	26.8%	1.139	34.297	-48.1%	1.322

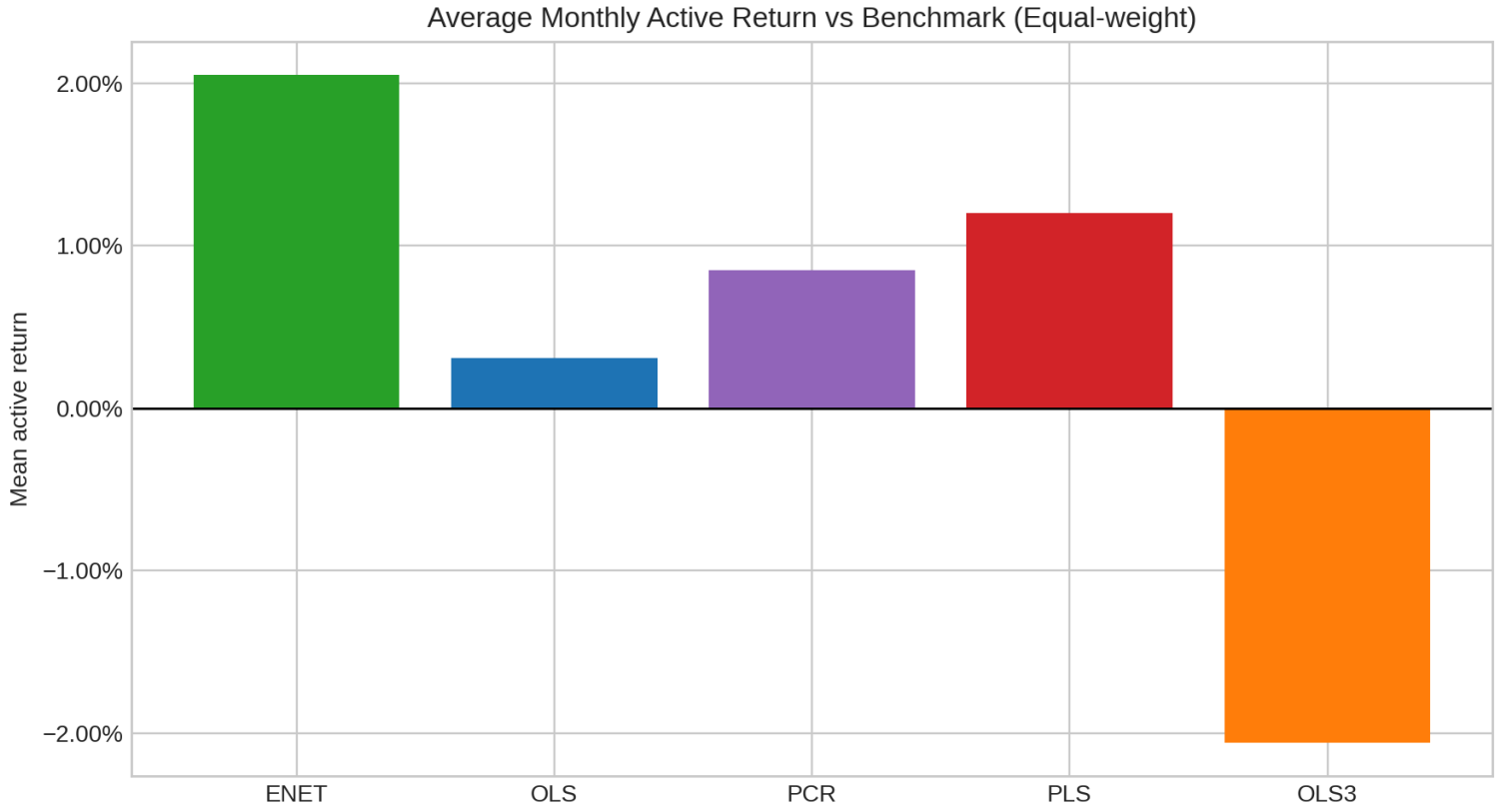
Model	Long Ann Ret	Long Sharpe	Long Cum Ret (x)	Long Max DD	LS Ann Ret	LS Sharpe	LS Cum Ret (x)	LS Max DD	LS Mean Turnover
OLS3	5.1%	0.325	1.114	-60.2%	-4.5%	-0.054	-0.498	-74.8%	1.055

Interpretation

- `ENET` is decisively best on the **long-short** portfolio dimension. Its equal-weight long-short strategy posts the highest annualized return, highest Sharpe, and strongest cumulative return after 30 bps cost.
- `OLS` is the best **long-only** model in this completed group by net Sharpe and annualized return.
- `PLS` and `PCR` remain investable in the sense that both produce long-short Sharpe ratios above 1.1 after cost, but neither matches `ENET`.
- `OLS3` is not competitive. Its long-short strategy is negative after cost and its drawdown is the worst of the completed set.

Benchmark Relative Results



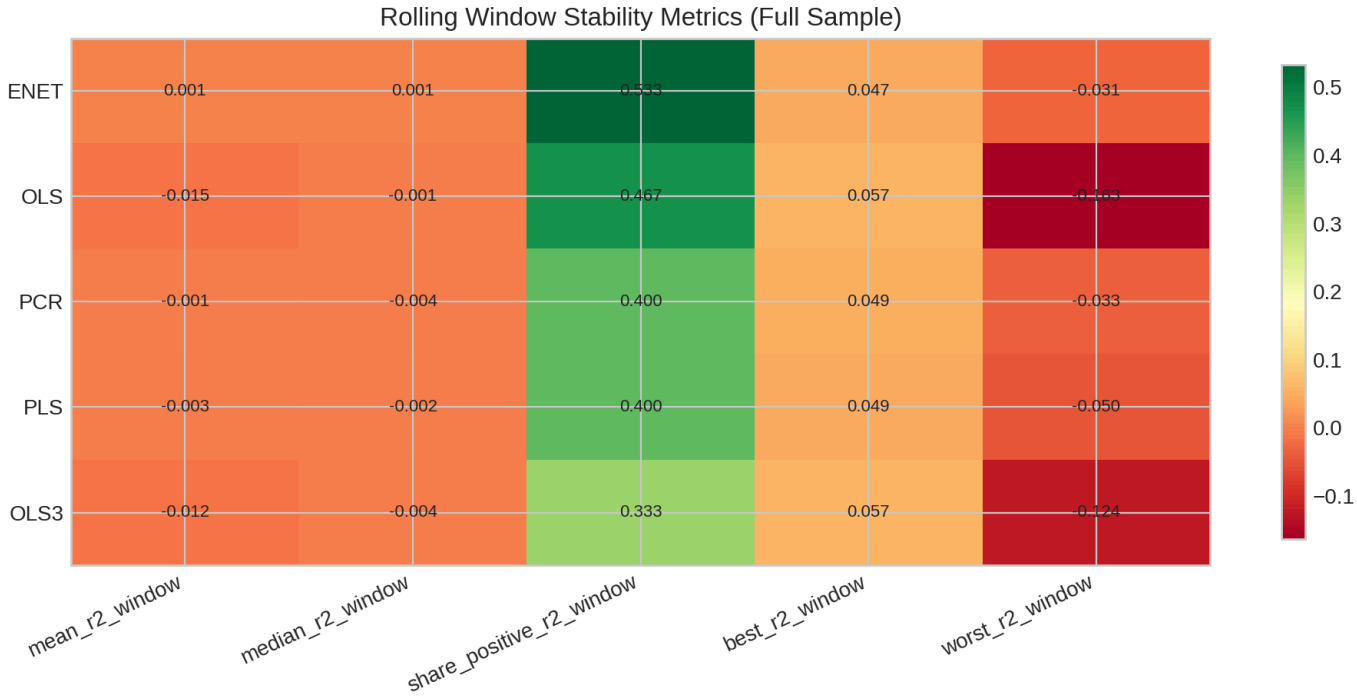


Model	IR vs EW Benchmark	Mean Active Return	Corr vs Benchmark	IR vs VW Benchmark
ENET	0.247	2.05%	0.182	0.329
OLS	0.034	0.31%	-0.043	0.023
PCR	0.098	0.85%	0.090	0.107
PLS	0.141	1.20%	0.098	0.156
OLS3	-0.244	-2.06%	0.074	-0.223

Interpretation

- `ENET` is the strongest benchmark-relative strategy by a clear margin.
- `PLS` and `PCR` are second tier but still positive.
- `OLS` is only slightly positive against the equal-weight benchmark.
- `OLS3` is negative against both equal-weight and value-weight benchmark comparisons.

Rolling Window Stability

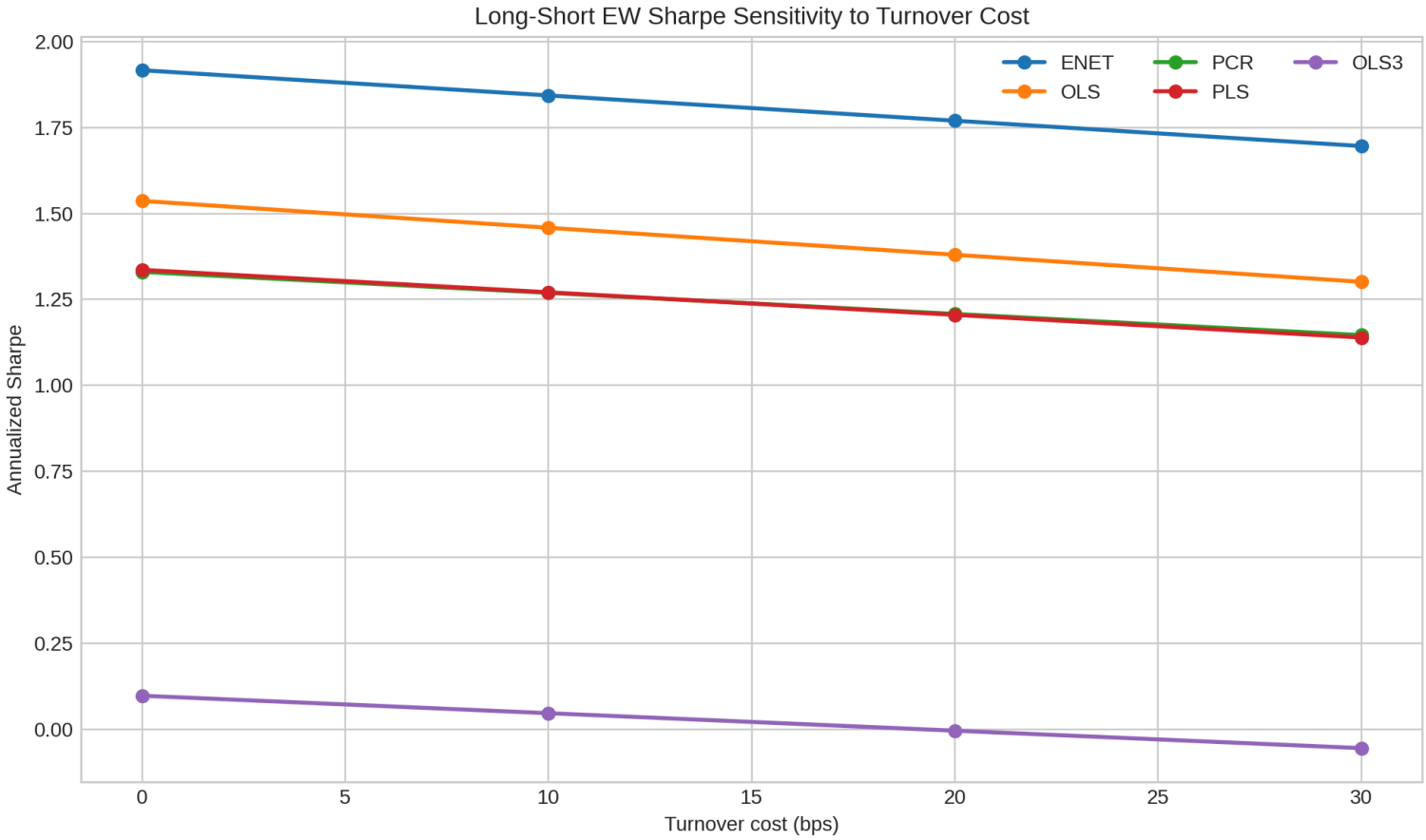


Model	Mean Window R2	Median Window R2	Share Positive Windows	Best Window R2	Worst Window R2
OLS	-0.0149	-0.0011	46.7%	0.0565	-0.1631
OLS3	-0.0115	-0.0038	33.3%	0.0574	-0.1242
ENET	0.0013	0.0009	53.3%	0.0467	-0.0307
PLS	-0.0030	-0.0021	40.0%	0.0486	-0.0499
PCR	-0.0007	-0.0043	40.0%	0.0492	-0.0326

Interpretation

- `ENET` is the only completed model with **positive mean full-sample window** **`R²`** and more than half of windows positive.
- `OLS` has the best single window but also much worse downside window risk than `ENET`. It is more fragile over time.
- `PCR` and `PLS` are middling and slightly negative on average window `R²`, but less erratic than `OLS`.
- `OLS3` is consistently weak.

Turnover Cost Sensitivity

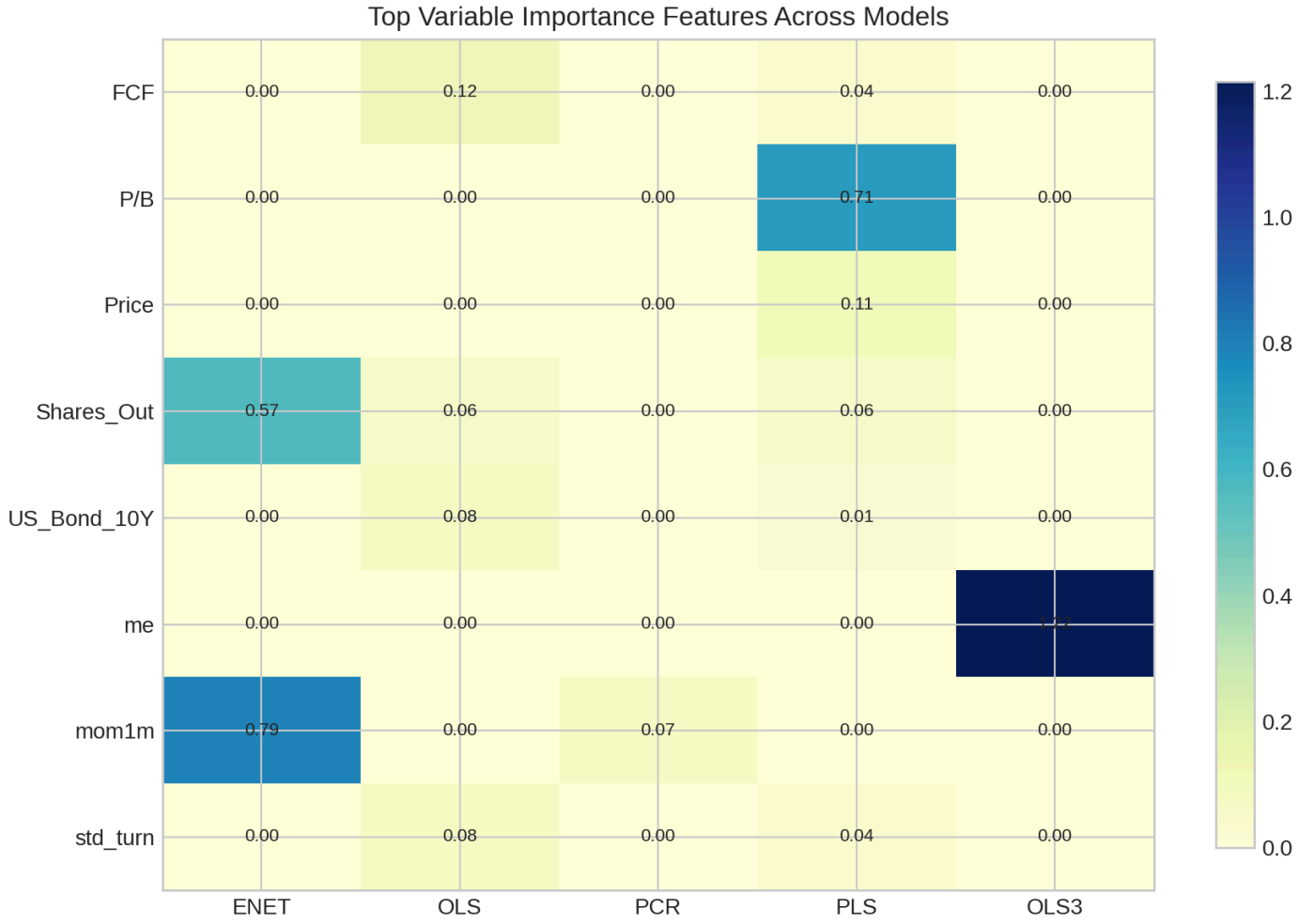


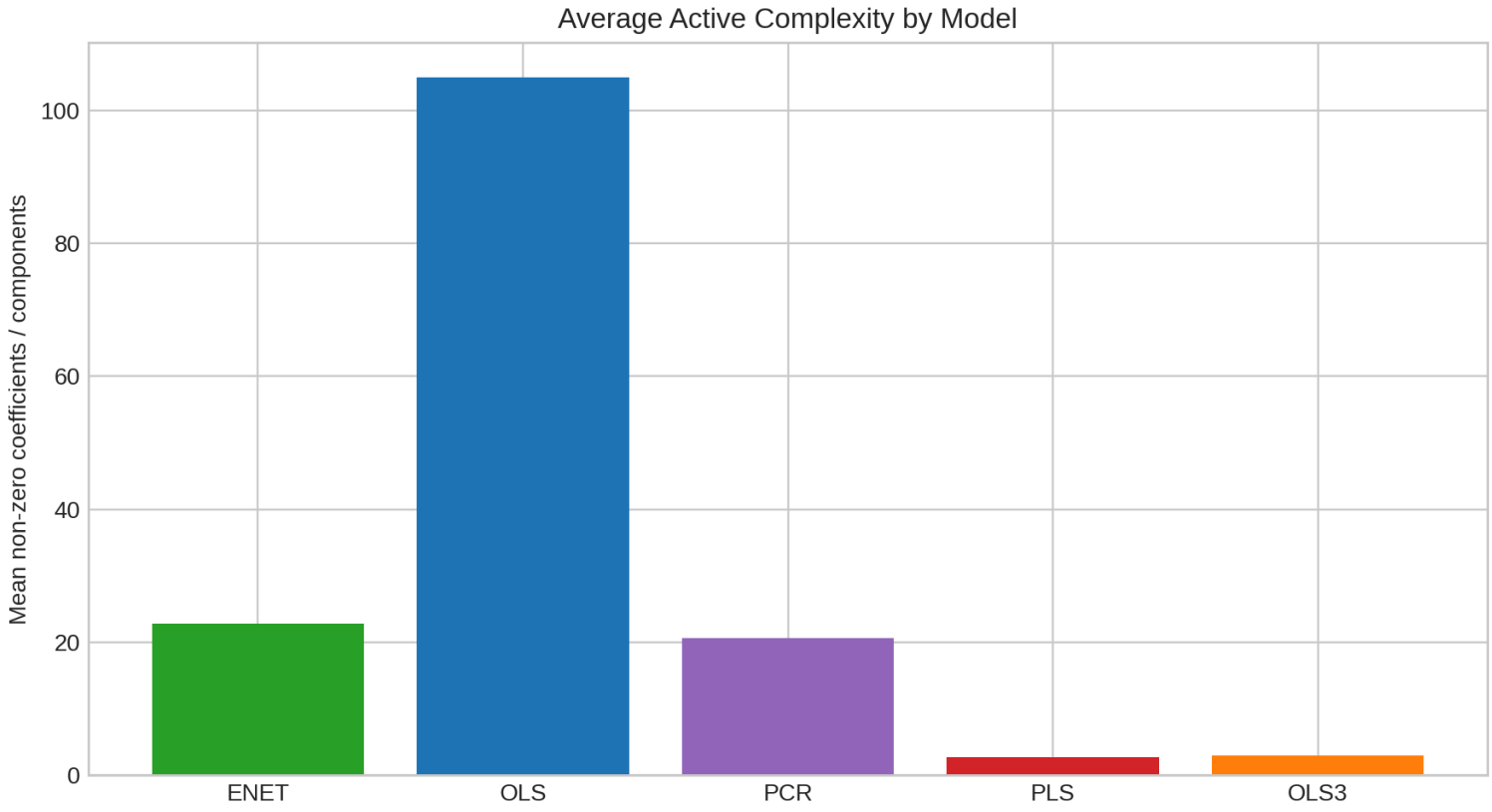
Model	LS Sharpe 0bps	LS Sharpe 10bps	LS Sharpe 20bps	LS Sharpe 30bps	Long Sharpe 30bps
OLS	1.536	1.458	1.380	1.301	1.265
OLS3	0.098	0.047	-0.003	-0.054	0.325
ENET	1.916	1.843	1.770	1.696	0.970
PLS	1.336	1.271	1.205	1.139	0.842
PCR	1.330	1.269	1.208	1.146	0.825

Interpretation

- `ENET` remains the best long-short model even after 30 bps turnover cost.
- `OLS` starts from a strong gross result but degrades faster than `ENET` once cost is applied.
- `PLS` and `PCR` are moderately resilient to cost, but still well below `ENET`.
- `OLS3` is effectively non-viable once cost is introduced.

Feature Importance and Model Shape





Top features by model

Model	Top 1	Top 2	Top 3	Top 4	Top 5
ENET	mom1m (0.789)	Shares_Out (0.566)	cfp (0.039)	ROE_Reported (0.000)	EBITDA (0.000)
OLS	FCF (0.120)	US_Bond_10Y (0.079)	std_turn (0.077)	mom6m (0.065)	Hong_Kong_Index (0.061)
PCR	Oper_CF (0.075)	agr (0.075)	dollar_vol (0.074)	Cur_Assets (0.074)	Inventory_BS (0.074)
PLS	P/B (0.709)	Price (0.107)	Shares_Out (0.061)	FCF (0.038)	std_turn (0.035)
OLS3	me (1.215)	be_me (-0.004)	ret_12_1 (-0.212)		

Complexity summary

Model	Mean Validation Score	Mean Active Complexity	Min Active Complexity	Max Active Complexity
OLS	0.298	105.000	105.000	105.000
OLS3	0.141	3.000	3.000	3.000
PLS	0.141	2.667	1.000	6.000
ENET	0.141	22.867	5.000	54.000
PCR	0.140	20.667	1.000	49.000

Extra Notes

How Feature Importance was calculated

1. First computes a baseline fit score on `base` using the same `R2OOS` function used elsewhere in the pipeline:

$$R^2_{\text{base}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i y_i^2}.$$

6. Then, for each feature (f), it sets that feature to `0.0` in `train`, `validation`, and `base`, refits the model, predicts again, and recomputes the same `R2OOS`.
7. The loss of fit from neutralizing that feature is:

$$\Delta R^2_f = R^2_{\text{base}} - R^2_{f=0}.$$

8. Finally, normalizes those losses into the reported variable-importance share:

$$\text{var_imp}_f = \frac{\Delta R^2_f}{\sum_j \Delta R^2_j}.$$

How Model Complexity was calculated

General logic

For each rolling window, the model returns:

- `best_params`: the hyperparameters chosen on the validation set,
- `best_score`: the validation objective value for the selected specification,
- `complexity`: a compact model-shape summary.

Complexity by model family

```
OLS / OLS3
```

These are implemented with `HuberRegressor`, not classical closed-form OLS.

- `best_score`: validation **RMSE**
- `n_nonzero_coef`: number of fitted coefficients with absolute value greater than zero after the final `train + validation` refit

Interpretation:

- For `OLS`, this is basically the number of active predictors in the robust linear fit.
- For `OLS3`, this stay at `3` because the model is fixed to `[me, be_me, ret_12_1]`.

ENET

- `best_score`: validation **RMSE**
- `best_alpha`: chosen penalty strength
- `best_l1_ratio`: chosen Elastic Net mixing weight
- `n_nonzero_coef`: number of coefficients that remain active after shrinkage

PLS

- `best_score`: validation **RMSE**
- `n_components`: chosen latent component count

PCR

- `best_score`: validation **RMSE**
- `n_components`: chosen principal-component count

GBRT

- `best_score`: validation **Huber loss**
- Complexity fields recorded:
  - `best_max_depth`
  - `best_min_samples_split`
  - `best_min_samples_leaf`
  - `best_learning_rate`

RF

- `best_score`: validation **MSE**
- Complexity fields recorded:
  - `best_max_depth`
  - `best_max_features